MIL-C-22931/10
10 February 1968
SUPERSEDING
MIL-C-22931/1A (in part)
27 March 1964

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC, .500 INCH OUTSIDE DIAMETER, 75 OHMS

The complete requirements for procuring cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: Copper wire or tube.
Overall diameter - 0.102 (2.59 mm) ± 0.001 (.03 mm) inch.

Dielectrics:
Cable core and insulating layer: Polystyrene.

Outer conductor: Aluminum tubing, smooth, seamless.
Inside nominal diameter: 0.421 inch (10.69 mm).
Outside diameter: 0.500 (12.70 mm) inch.

Jacket: See table.

Voltage withstand:
1,800 volts rms (nominal), or
3,200 volts dc.

Insulation defects, spark: See table.

Attenuation (decibels (dB) per 100 ft maximum):
0.44 dB at 30 MHz.
1.70 dB at 400 MHz.
2.80 dB at 1,000 MHz.

Velocity: 87.5 percent, nominal.

Capacitance: 14.7 pf/ft, nominal.

Impedance: 75 ±2 ohms.

Voltage standing wave ratio:
Initial: 1.15 maximum for 10-1,000 MHz.
Following bending and temperature cycling: 1.18 maximum for 10-1,000 MHz.

Mandrel size for temperature cycling, bending, and cold bend tests: 15 inches (381.0 mm).

Minimum bending radius: 5 inches (127.0 mm).

FSC 6145

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Requirement 1/	M22931/10 dash number	
	-001	-002
Jacket	Not applicable	Applicable
Diameter +0.020(.51) -0.010(.25)		0.575(14.61)
Minimum wall thickness		0.035(.89)
Insulation defects, spark	Not applicable	Applicable

1/ Dimensions are in inches; millimeters are in parentheses. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

NOTES:

1. Cross reference information is as follows:

Part number	RG number	Former classification Class	Former classification Type
M22931/10-001	RG244/U	N	I
M22931/10-002	RG245/U		

2. Figure is for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

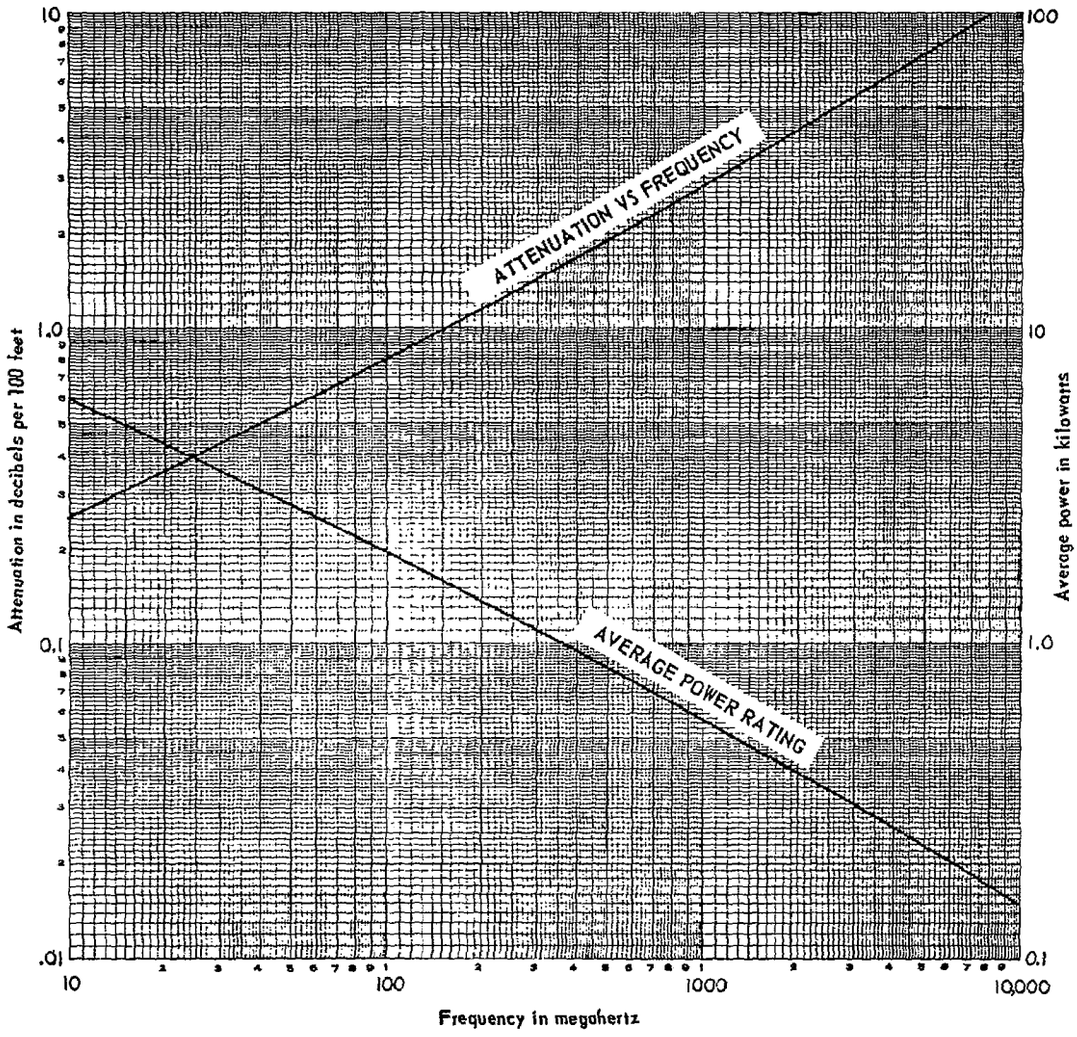
Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - AS, OS MC
Air Force - 19

Preparing activity:
Navy - EC

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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. *Attenuation and power rating curves.*

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